

ConSight — Feature Analysis & Recommendation Rationale

Why These Features, Why This Priority, Why Not Others

Document Purpose

This document captures the complete reasoning behind the recommended add-on features for ConSight (SIH26122). It explains:

- What the problem statement actually demands
- What the competitive landscape already offers
- Why certain features were recommended over the original Phase 5-12 roadmap
- Why specific features were prioritized or dropped
- What really matters for winning SIH26122

1. Understanding the Problem Statement (SIH26122)

What Oil India Limited Actually Asked For

The problem statement from Oil India Limited (SIH26122) has five explicit requirements:

Requirement 1 — Heterogeneous Input Ingestion

"Ingest heterogeneous discipline-wise inputs — free-text daily reports, spreadsheets, scanned diaries, Primavera/MS Project exports — and extract activity-level actual start/end events."

Translation: ConSight must accept multiple input formats AND connect to enterprise scheduling tools (Primavera P6, MS Project). This is not optional — it is the core of the problem.

Requirement 2 — LLM-Based Conversational Interface

"Offer an LLM-based conversational or voice interface ('time agent') for site supervisors across disciplines to log activity start/end with minimal friction."

Translation: A text-based Time Agent satisfies "conversational." Voice is a bonus, not a requirement. The key phrase is "minimal friction" — which means the input channel must be as easy as possible for supervisors.

Requirement 3 — Fuzzy Matching to L5/L6 Plan

"Fuzzy-match and link extracted discipline-specific activity descriptions to the correct L5/L6 plan node, handling terminology differences and granularity mismatches, and flag unmatched/new activities for planner review rather than silently dropping them."

Translation: The matching engine must handle terminology gaps and must not drop unmatched work. The planner review workflow is explicitly required.

Requirement 4 — Confidence Score and Audit Trail

"Auto-update actual start/end dates in the schedule/PMIS in near real time, with a confidence score and audit trail per entry."

Translation: Two things — (a) actual dates must flow back to the schedule/PMIS, and (b) every entry needs confidence + audit. The phrase "auto-update actual start/end dates in the schedule/PMIS" means the system must write back to the enterprise schedule, not just display internally.

Requirement 5 — Institutional Memory

"Produce a clean, structured, discipline-tagged actual-progress dataset that serves two purposes: (a) live input for performance analytics, delay/risk pattern discovery, and forecasting, and (b) a foundation for institutional memory building — a growing, queryable repository of real project execution patterns."

Translation: The system must produce structured data that grows over time and becomes queryable institutional memory. This is not just a dashboard — it is a knowledge base.

What This Means for Feature Selection

Every recommended feature must map to one of these five requirements. Features that don't map to any requirement are scope creep and should be deferred.

2. The Competitive Landscape — What Already Exists

Why This Matters

Before recommending features, it is critical to understand what the market already does. If a feature already exists in 5+ products, recommending it as a "differentiator" would be dishonest. The goal is to identify the gap — what no competitor offers — and fill it.

Category A: Field Capture Tools (WhatsApp-Based)

These tools capture site updates via WhatsApp and generate reports.

Tool	WhatsApp	AI Extraction	Multi-Language	Matching to Schedule	Confidence	Planner Review	Audit Trail
Banamind	Yes	Yes	Any language	No	No	No	Partial

Tool	WhatsApp	AI Extraction	Multi-Language	Matching to Schedule	Confidence	Planner Review	Audit Trail
Velora AI	Yes	Yes	Hinglish, Telugu	No	No	No	Partial
Updatly	Yes	Yes	No	No	No	No	No
BrickVox	Yes (voice)	Yes	6+ languages	No	No	No	No
SiteSetu	Yes	Yes	Hindi-English	No	No	No	No
Miru Build	Yes	Yes	No	No	No	No	No
Aasaan	Yes	Partial	No	No	No	No	No

Key insight: These tools stop at report generation. None of them match field updates to L5/L6 schedule activities. None have confidence scoring or planner review.

Category B: Schedule Intelligence Tools

These tools analyze Primavera P6 / MS Project schedules for delays and critical path impact.

Tool	P6 Import	Delay Analysis	Critical Path	Downstream Impact	Field Input	AI Extraction	Confidence	Planner Review
SmartPM	Yes	Yes	Yes	Yes	No	No	No	No
CritPath	Yes	Yes	Yes	Yes	No	No	No	No
Verian	Yes (native)	Yes	Yes	Yes	No	No	No	No
Planera Manny	Yes	Yes	Yes	Yes (what-if)	No	No	No	No
PlanLens	Yes	Yes (forensic)	Yes	Yes	No	No	No	No
AskSURI	Yes	Yes	Yes	Yes	No	No	No	No

Key insight: These tools analyze schedules in isolation. None of them ingest field-reported progress. None connect what supervisors report to what the schedule says. They are retrospective, not real-time.

Category C: Enterprise Project Management

Tool	Field Capture	Schedule	Photo Evidence	AI Features	India-Specific	Pricing
Procore	Manual forms	P6/MS Project	Yes	Limited	No	Enterprise (\$\$\$)
Autodesk Build	Manual forms	P6/MS Project	Yes	Partial	No	Enterprise (\$\$\$)

Tool	Field Capture	Schedule	Photo Evidence	AI Features	India-Specific	Pricing
InEight	Mobile capture	P6	Yes	Partial	No	Enterprise (\$\$\$)
OpenSpace	360 camera	P6/MS Project	Visual AI	Yes	No	Enterprise (\$\$\$)
Buildots	Helmet camera	BIM compare	Yes	Yes	No	Enterprise (\$\$\$)

Key insight: These are expensive, enterprise-grade tools with manual form-based capture. None are designed for Indian site conditions. None use AI to extract structured progress from free text. None have confidence-based human-in-the-loop matching.

Category D: Indian Construction Apps

Tool	WhatsApp	Multi-Language	Offline	DPR	Schedule Matching	AI	Confidence
Onsite	No	8 languages	Yes	Yes	No	No	No
Yojo	No	12 languages	Yes	Yes	No	No	No
Powerplay	No	Hindi	No	Yes	No	No	No
Buildary	No	8 languages	No	Yes	No	No	No

Key insight: Indian apps understand language and offline needs but have no AI extraction, no schedule matching, no confidence scoring, and no planner review workflow.

The Gap

No single tool in the market offers the complete pipeline:

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Field Capture (WhatsApp/multi-language)
  → AI Extraction
    → Hybrid Matching to L5/L6 Plan
      → Confidence Scoring
        → Planner Review (approve/correct/reject/create new)
          → Audit Trail
            → Schedule Update (P6 export)
              → Delay Ripple Analysis
                → Institutional Memory

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Field capture tools have the left side but not the right.

Schedule intelligence tools have the right side but not the left.

Enterprise PM tools have the middle but are manual, expensive, and not India-specific.

Indian apps have language and offline but no AI intelligence layer.

ConSight fills the gap by connecting both ends with AI and human oversight.

3. Analysis of the Original Phase 5-12 Roadmap

What the Roadmap Proposed

Phase	Feature	Assessment
Phase 5	Voice Time Agent	Good alignment — PS says "conversational or voice." But text Time Agent already satisfies the requirement. Voice is a bonus.
Phase 6	OCR & Document Intelligence	Good alignment — PS mentions "scanned diaries." But OCR is unreliable in live demos. Pitch as roadmap, don't demo.
Phase 7	Delay & Risk Prediction	Good alignment — PS mentions "delay/risk pattern discovery." But should be simpler than a full ML model — variance analysis from audit data is sufficient for SIH.
Phase 8	Weather & External Context	Not in problem statement. Drop for SIH.
Phase 9	Advanced RAG / Institutional Memory	PS mentions "institutional memory." But Phase 4 Insights page already starts this. "Advanced RAG" is overkill for SIH — queryable audit data is enough.
Phase 10	Mobile / Site Application	PS implies field use. React PWA or WhatsApp covers this. Don't build a separate mobile app.
Phase 11	BIM / Digital Twin	Not in problem statement. Way too complex. Drop entirely.
Phase 12	Multi-Project / Enterprise	Not in problem statement. Single project demo is sufficient. Drop for SIH.

Critical Gap: Primavera P6 / MS Project Round-Trip

The original roadmap completely misses this. The problem statement explicitly requires:

1. **Import:** "Ingest heterogeneous discipline-wise inputs — ... Primavera/MS Project exports"
2. **Export:** "Auto-update actual start/end dates in the schedule/PMIS in near real time"

Oil India Limited uses Primavera P6. If the demo shows ConSight capturing field progress but cannot import or export P6 schedule data, the OIL judges will ask: "How does this connect to our existing Primavera setup?" The roadmap has no answer.

This is not a future feature — it is a core requirement that should be the first thing built after Phase 4.

Critical Gap: WhatsApp Integration

The problem statement says "minimal friction" and "low-friction mechanism." In India, the lowest-friction communication channel is WhatsApp (530 million active users in India). The original roadmap has Voice (Phase 5) and Mobile (Phase 10), but both require the supervisor to install something new. WhatsApp requires nothing — they already use it daily.

Critical Gap: Multi-Language Input

Indian site reports mix English with Hindi, Tamil, Telugu, and regional languages. "XX-101 ka spool erection 9:30 baje start kiya" is how supervisors actually speak. The original roadmap doesn't mention multi-language support at all. Without it, the AI extraction fails on real Indian site input.

Critical Gap: Delay Ripple (Real-Time, Not Predictive)

The original roadmap has "Delay & Risk Prediction" (Phase 7), which is about predicting future delays using ML. This is harder to build and harder to demo. What the PS actually needs is simpler and more immediate: when a delay event is logged, show which downstream activities are affected right now. This is a graph traversal on existing dependency data, not a prediction model.

4. Recommended Features — Reasoning for Each

Feature 1: Primavera P6 / MS Project Round-Trip

Why it matters: Directly required by the problem statement. Oil India Limited uses Primavera P6. Without this, ConSight is an island — it captures field progress but can't connect to the enterprise schedule.

Why it's priority 1: The PS asks for it explicitly in two places (import and export). No other feature has this level of direct requirement.

Why it's not in the original roadmap: Unclear. Possibly assumed that the schedule would be manually entered. But the PS says "Primavera/MS Project exports" — it expects the system to ingest these formats.

Market context: Shape, Outbuild, OpenSpace, CritPath, PlanLens, Verian all do this. It is a solved technical problem. ConSight is not inventing anything new here — it is meeting a baseline requirement.

Implementation: Use xerparser for P6 .xer files and mpxj or aspose-tasks for MS Project .mpp files. Map P6 activity fields to ConSight's schema. Add export endpoint that writes actual dates back to P6 format.

Effort: Medium (4-5 days)

Feature 2: WhatsApp Integration

Why it matters: The PS says "minimal friction" and "low-friction mechanism." WhatsApp is THE low-friction channel in India. 530 million users. Supervisors already use it daily. No new app to install, no new interface to learn.

Why it's priority 2: Highest demo impact. Sending a WhatsApp message and watching it appear on the dashboard with a match is visually powerful. It directly demonstrates "low-friction" to

the judges.

Why over Voice (original Phase 5): Voice requires speech-to-text, which is error-prone, especially with Indian accents and code-mixed languages. WhatsApp text messages are more reliable, more demoable, and reach more supervisors. Voice can be a future enhancement — WhatsApp satisfies the "conversational" requirement.

Why over Mobile App (original Phase 10): A separate mobile app requires installation, onboarding, and training. WhatsApp requires none of that. The React PWA can serve as the planner's interface; WhatsApp serves as the supervisor's interface.

Market context: Banamind, Velora, Updatly, BrickVox, SiteSetu all do WhatsApp-based reporting. But none of them connect WhatsApp input to L5/L6 schedule matching with confidence scoring. ConSight's WhatsApp → extraction → matching → confidence → planner review pipeline is unique.

Implementation: WhatsApp Business Cloud API webhook in FastAPI. Reuse existing `agent_service.chat()`. The WhatsApp endpoint is just another input channel — same extraction, same matching, same confidence engine.

Effort: Medium (3-4 days)

Feature 3: Multi-Language / Hinglish Extraction

Why it matters: Indian site supervisors don't speak pure English. They use Hindi-English mixed language (Hinglish), Tamil-English, Telugu-English. If the AI extraction fails on "XX-101 ka spool erection start kiya," the system is useless on real Indian sites.

Why it's priority 3: Lowest effort (1-2 days of prompt engineering), highest emotional impact in a demo. When a judge sees Hinglish input being correctly extracted and matched, it immediately signals "this was built for India, not adapted for India."

Why it's not in the original roadmap: The original roadmap focuses on voice (Phase 5) and OCR (Phase 6) as input methods but doesn't address the language of the input. Multi-language is more fundamental — it affects every input channel (text, WhatsApp, Time Agent, even Excel descriptions).

Market context: Velora AI handles Hinglish/Telugu. BrickVox supports 6+ languages. SiteSetu reads mixed Hindi-English. But these are field-capture tools — they structure the message but don't match it to a schedule. ConSight's multi-language extraction → hybrid matching to English schedule activities is unique.

Implementation: Update the extraction service's system prompt to handle code-mixed input. Add few-shot examples in the prompt. Test with a benchmark set of 20-30 code-mixed messages. No new architecture — just prompt engineering.

Effort: Low (1-2 days)

Feature 4: Delay Ripple / Critical Path Impact

Why it matters: The PS asks for "delay/risk pattern discovery." But the original roadmap's "Delay & Risk Prediction" (Phase 7) implies a predictive ML model, which is hard to build, hard to validate, and hard to demo reliably.

What's more practical and more impressive is real-time delay ripple: when a supervisor logs a delay event, ConSight immediately shows which downstream activities are impacted, by how many days, and whether the critical path is affected. This is deterministic graph traversal, not prediction — it always works, it's always correct, and it's visually powerful.

Why it's priority 4: It connects field events to schedule consequences. Most tools do one or the other. ConSight showing "your delay on PIP-1023 will push PIP-1024 by 2 days and put the critical path at risk" is a moment that judges remember.

Why over the original Phase 7: Prediction models need training data, validation, and confidence intervals. In a hackathon, you don't have time to train a reliable model. Delay ripple is deterministic — it uses the dependency graph that already exists in the P6 import. It always works.

Market context: SmartPM, CritPath, Verian, Planera Manny all do delay impact analysis. But they analyze schedule files in isolation — they don't connect field-reported progress to schedule impact in real time. ConSight's field delay → immediate schedule ripple is unique.

Dependency: Requires Feature 1 (P6 round-trip) because it needs predecessor/successor data from the schedule.

Implementation: Store activity relationships (predecessors, successors, relationship type, lag) from P6 import. When a DELAY event is logged and matched, traverse the successor chain. Calculate delay propagation based on relationship type (FS, SS, FF, SF) and available float. Display on dashboard and Insights page.

Effort: Medium (3-4 days)

Feature 5: Productivity & Pattern Analytics (Institutional Memory Deep Dive)

Why it matters: The PS specifically asks for "a foundation for institutional memory building — a growing, queryable repository of real project execution patterns (actual durations, recurring delay causes, discipline-wise productivity)." Phase 4 Insights page starts this, but it needs depth to be genuinely useful and to differentiate from a regular dashboard.

Why it's priority 5: The institutional memory concept is the PS's differentiator — it's what makes this more than a progress tracking tool. It's the "growing, queryable" part that judges will probe. "Does this actually accumulate knowledge over time?"

Why over the original Phase 9 (Advanced RAG): RAG implies vector databases, embedding search, and natural language querying of unstructured knowledge. This is overkill for SIH. What the PS actually asks for is structured, queryable data — actual durations, delay patterns, discipline productivity. This is SQL queries on the existing audit trail, not a RAG system.

Market context: SmartPM does schedule performance analytics. InEight does earned value. But none frame it as "institutional memory" — a knowledge base that grows across projects. ConSight's framing of audit data as institutional memory is unique.

Implementation: Add SQL queries that aggregate audit trail data: actual vs planned duration by discipline, recurring delay patterns by location, supervisor productivity index, variance trend over time, activity type performance. Display on Insights page with charts.

Effort: Low (2-3 days — mostly SQL + frontend charts)

5. Features Dropped From the Original Roadmap — Reasoning

Phase 8: Weather & External Context

Why dropped: Not mentioned in the problem statement. Weather is a nice contextual input but doesn't directly address any of the five PS requirements. Building weather integration adds complexity (external API, location mapping, data freshness) without adding demo value.

If needed later: Can be added as a context field on progress events. But not for SIH.

Phase 11: BIM / Digital Twin Integration

Why dropped: Not in the problem statement. BIM integration requires IFC model files, 3D visualization infrastructure, and model-element-to-activity mapping. This is a separate product, not a feature. It would take weeks to build and would be unreliable in a demo.

If needed later: Explore as a post-SIH research direction. Not practical for a hackathon.

Phase 12: Multi-Project / Enterprise Layer

Why dropped: Not in the problem statement. The PS asks for a solution for a single infrastructure project. Multi-project support adds database schema changes, role-based access control, organization management, and cross-project analytics. None of this is needed for the SIH demo.

If needed later: Add project_id to all tables, add role-based middleware. But not for SIH.

Phase 10: Separate Mobile Application

Why dropped: The React frontend can be built as a responsive PWA that works on mobile. WhatsApp integration covers the supervisor's mobile use case. A separate native mobile app (Flutter/React Native) is unnecessary complexity.

If needed later: PWA with offline support is sufficient. Native app only if push notifications or device hardware (camera, GPS) are needed.

6. Priority Ranking — Reasoning

How Priority Was Determined

Three factors:

1. **PS alignment** — Does the feature directly address a problem statement requirement?
2. **Judge impact** — Will this feature create a memorable demo moment?
3. **Effort** — Can it be built and tested within the available time?

Priority 1: P6/MS Project Round-Trip

- PS alignment: Direct — explicitly mentioned twice (import and export)
- Judge impact: Critical — without this, OIL judges can't see how ConSight fits their workflow
- Effort: Medium (4-5 days)
- Verdict: Must have. Not building this is a disqualifying gap.

Priority 2: WhatsApp Integration

- PS alignment: Direct — "minimal friction" and "low-friction mechanism"
- Judge impact: Very high — visually powerful demo moment
- Effort: Medium (3-4 days)
- Verdict: Must have for demo impact. The single most impressive thing to show judges.

Priority 3: Multi-Language / Hinglish Extraction

- PS alignment: Indirect — "across disciplines" implies different languages, Indian context
- Judge impact: High — emotional resonance, shows India-specific design
- Effort: Low (1-2 days)
- Verdict: Quick win. Highest ROI — 1 day of work for a moment judges will remember.

Priority 4: Delay Ripple / Critical Path Impact

- PS alignment: Direct — "delay/risk pattern discovery"
- Judge impact: High — shows system understands schedule consequences
- Effort: Medium (3-4 days)
- Verdict: Strong feature. Depends on P6 import for dependency data.

Priority 5: Productivity Analytics (Institutional Memory)

- PS alignment: Direct — "institutional memory building"
- Judge impact: High — shows the system produces lasting value, not just real-time matching
- Effort: Low (2-3 days)
- Verdict: Builds on existing Phase 4 Insights page. Completes the "institutional memory" story.

7. What Really Matters for Winning SIH26122

The Three Things That Win

1. A flawless Phase 1-4 demo.

The 3-case demo (clean match, ambiguous correction, unmatched create-new) must work perfectly every time. A smooth demo with no bugs beats 20 features that crash. Practice until the demo is muscle memory. Have a mock/fallback mode ready in case the live API fails.

2. P6/MS Project round-trip.

This is the feature that separates "student project" from "deployable solution." When OIL judges see ConSight import their Primavera P6 schedule, match field progress to it, and export updated actuals back, they see a tool that fits their existing workflow. Without this, ConSight is an interesting prototype. With it, ConSight is a potential deployment.

3. The Hinglish WhatsApp moment.

When a judge sees a WhatsApp message in Hindi-English mixed language — "XX-101 ka spool erection 9:30 baje start kiya" — being extracted, matched to the correct L5/L6 activity, and auto-confirmed at 94% confidence, they understand three things instantly:

- This was built for India, not adapted for India
- This requires zero new software for supervisors
- This actually works on real Indian sites

That 10-second moment is worth more than any feature list.

What Does NOT Matter for Winning

- BIM / Digital Twin — Out of scope, not in PS, too complex
- Multi-project / Enterprise — Not needed for SIH demo
- Weather integration — Not in PS
- Voice processing — Text Time Agent satisfies the PS requirement
- OCR — Mention as roadmap, don't demo (unreliable)
- Advanced RAG — Queryable audit data is sufficient for "institutional memory"

- Native mobile app — PWA + WhatsApp covers this

The Pitch in One Sentence

ConSight captures field progress from WhatsApp in Hindi, AI-matches it to the L5/L6 Primavera schedule with 94% confidence, auto-updates actuals, shows downstream delay impact, and builds institutional memory — with a human-in-the-loop audit trail on every decision.

8. Merging Both Documents

Take From the Original Roadmap

- Phase 5 (Voice) — mention as future enhancement
- Phase 6 (OCR) — mention as future enhancement
- Phase 7 (Delay Prediction) — simplify to delay ripple (deterministic, not predictive)
- Phase 9 (RAG) — simplify to structured analytics on audit data

Take From the Recommended Features

- P6/MS Project Round-Trip — make this Phase 5 (highest priority)
- WhatsApp Integration — make this Phase 6
- Multi-Language Extraction — make this Phase 7 (quick win)
- Delay Ripple — make this Phase 8 (depends on P6 import)
- Productivity Analytics — make this Phase 9 (builds on Insights)

Drop From Both

- Phase 8 (Weather) — not in PS
- Phase 11 (BIM) — not in PS, too complex
- Phase 12 (Enterprise) — not in PS
- Phase 10 (Mobile app) — PWA + WhatsApp covers this

Final Merged Roadmap

Phase	Feature	PS Alignment	Effort	Demo Impact
1-4	Core MVP (complete)	All requirements	Done	Foundation
5	P6/MS Project Round-Trip	Direct — import + export	Medium	Critical
6	WhatsApp Integration	Direct — "minimal friction"	Medium	Very high
7	Multi-Language / Hinglish	India-specific	Low	High
8	Delay Ripple / Critical Path	Direct — "delay/risk"	Medium	High

Phase	Feature	PS Alignment	Effort	Demo Impact
9	Productivity Analytics	Direct — "institutional memory"	Low	High
Future	Voice, OCR, RAG, Mobile PWA	Roadmap only	—	Mention in pitch

9. ConSight's Unique Value Proposition — Final Statement

Individual features exist in the market. WhatsApp reporting exists (Banamind, Velora). P6 import exists (Shape, OpenSpace). Delay analysis exists (SmartPM, CritPath). Multi-language exists (BrickVox, Yojo). Photo evidence exists (Procore, Onsite).

But no single platform connects field capture to schedule update with AI matching, confidence scoring, human-in-the-loop review, audit trail, delay ripple, and institutional memory.

ConSight's value is not in any single feature. It is in the **complete pipeline** — from a supervisor's WhatsApp message in Hindi to an updated Primavera P6 schedule with confidence, audit trail, and downstream delay impact.

That pipeline is what wins SIH26122.